BimmerStein ECU Tool

BMW MS41 Programming, Diagnostics, and Recovery

USER MANUAL

Version 0.1.0b5 - Windows x64

OFF-ROAD USE ONLY

Read the safety section before connecting to an ECU. Use stable power, confirm the selected operation, and never interrupt an active flash write.

Safety first

Read this section before connecting to an ECU or opening a write workflow.

IMPORTANT: OFF-ROAD, COMPETITION, RESEARCH, AND BENCH USE ONLY. Do not use this software to modify a vehicle operated on public roads. The user is responsible for compliance with applicable emissions, safety, registration, and other laws.

- Use a stable, regulated vehicle or bench power supply.
- Confirm the selected COM port, wiring mode, ECU variant, flash-chip family, and file size.
- Keep ignition ON and maintain power throughout an active write.
- Do not disconnect the FTDI adapter while a flash operation is running.
- A backup is recommended, but the Flash tab respects the operator's backup selection.
- Enable read-back verification when an independent byte comparison is required.
- Keep hardware BSL recovery available when testing boot-region or experimental firmware changes.

DANGER: If a write fails after erase has started, do not turn ignition off. Do not disconnect the adapter and do not close the application. Use **Retry Flash Recovery** while the retained high-speed or RAM-agent session is still active.

WARNING: Hardware BSL and armed Soft-BSL boot-region writes can rewrite code required for normal ECU startup. A failed boot-region write can leave the ECU unbootable until hardware BSL recovery.

After a successful flash

Every successful Flash-tab write ends with the same operator instruction:

1. Turn ignition OFF.
2. Wait at least 10 seconds.
3. Turn ignition ON.

The ignition cycle is an operational handoff, not a substitute for flash finalization or optional read-back verification.

Supported scope

Area	Supported scope	Important boundary
ECU families	BMW MS41.1, MS41.2, and MS41.3	MS41.0 identification, analysis, and selected workflows remain explicitly guarded where untested.
Normal communications	BMW DS2 over K-Line, 9600 baud, 8E2	DS2 is not KWP2000.
Stock fast transfers	Native-fast DS2 through an FTDI D2XX connection	The ECU-exact requested high rate is 187,500 baud.
Soft-BSL	Persistent loader plus RAM agents	Installation modifies firmware and requires the guided workflow.
Hardware BSL	Intel 28F200 and AMD/JEDEC 29F200/29F400	Uses a separate direct ASC0 tap, not the normal K-Line connection.
File sizes	256 KB full ROM and 24 KB tune	Choose the operation matching the file and intended region.
Host platform	Windows x64 installer or portable release	The regular PyInstaller package is recommended. Assets labeled `Nuitka-Experimental` are a second compatibility-testing option.

Flash-chip families

- **Intel 28F200** uses the Intel command set. Hardware-BSL erase/program operations require the correct external 12 V VPP/RP# supply.
- **AMD/JEDEC 29F200 and 29F400** use the AMD command set and are single-supply devices. Do not apply Intel VPP requirements to them.
- A 29F400 operation must also select the correct visible upper or lower half.

Install and start

Windows installer

1. Download the regular versioned `Windows-x64-Setup.exe` installer (recommended).
2. Run it for a per-user installation with no administrator access required.
3. Install the driver for the intended FTDI adapter, then launch **BimmerStein ECU Tool**.

Assets whose names contain `Nuitka-Experimental` are an explicitly **experimental** second option. That installer uses a distinct product identity and installation directory so it can coexist with the regular build for compatibility testing. Report the selected backend when describing startup or packaging behavior.

Portable Windows package

1. Verify the release ZIP checksum against its matching `.zip.sha256` file when supplied.
2. Extract the complete ZIP into a writable folder.
3. Run `BimmerStein ECU Tool.exe` from the extracted application folder.

Do not move only the executable. PyQt, protocol resources, patch descriptors, and RAM-agent payloads are stored under `_internal` in the regular package and beside the executable in the flat experimental Nuitka package. Keep the selected package's complete extracted folder together.

The executable may not be code-signed. Windows can show an unknown-publisher warning. Confirm the release filename and matching `.zip.sha256` value before continuing.

Interface driver

Install the driver for the selected FTDI adapter. D2XX is preferred for stock native-fast DS2, Soft-BSL, and hardware BSL. When D2XX is unavailable, normal DS2 and supported hardware-BSL paths can fall back to pyserial at their compatible rates.

Data folders

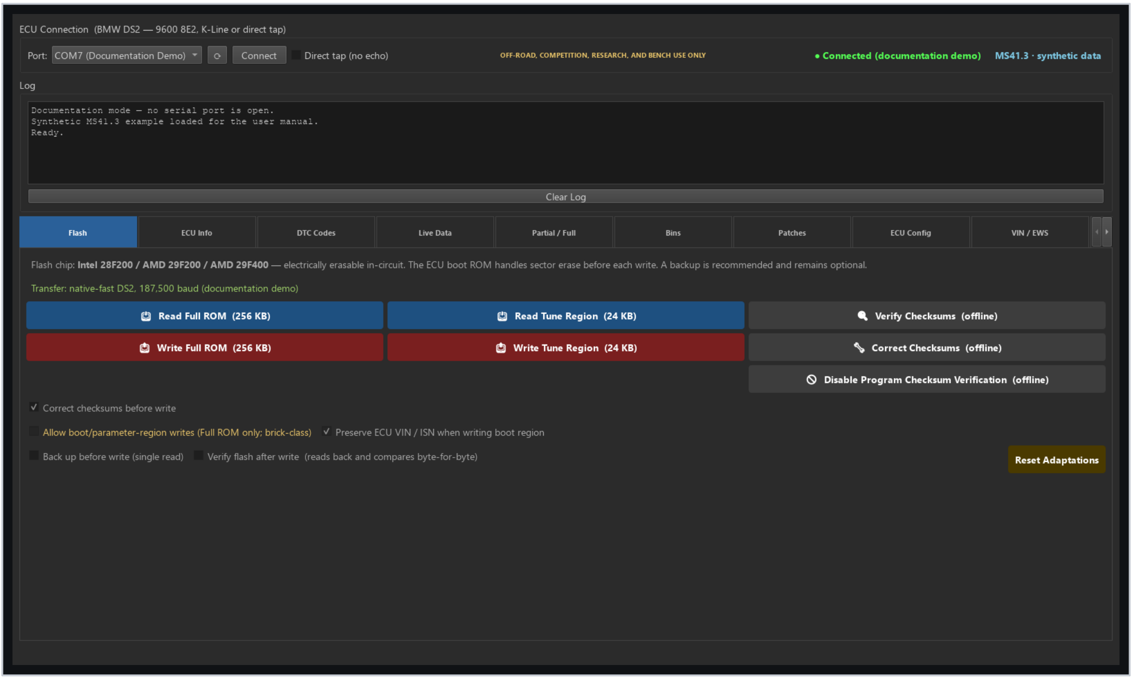
The portable application creates mutable data beside the executable:

- `backups/` for saved reads, generated images, and recovery material.
- `logs/` for session logs and diagnostic detail.

The ROM Analyzer stores user-imported calibration definitions under `%LOCALAPPDATA%\BimmerStein ECU Tool\definitions\`. This keeps the selected definition available when the portable application folder is replaced during an update. No calibration definition is included with the application.

Treat full ROMs and logs as private. They can contain a VIN, calibration identity, ECU identity, and operation history.

Main window



Main Flash workspace while disconnected

The connection bar remains visible above all tabs. It contains the normal DS2 COM selection, Connect button, direct-tap choice, connection state, ECU variant, and transfer-mode information. The shared log and progress controls remain visible below it.

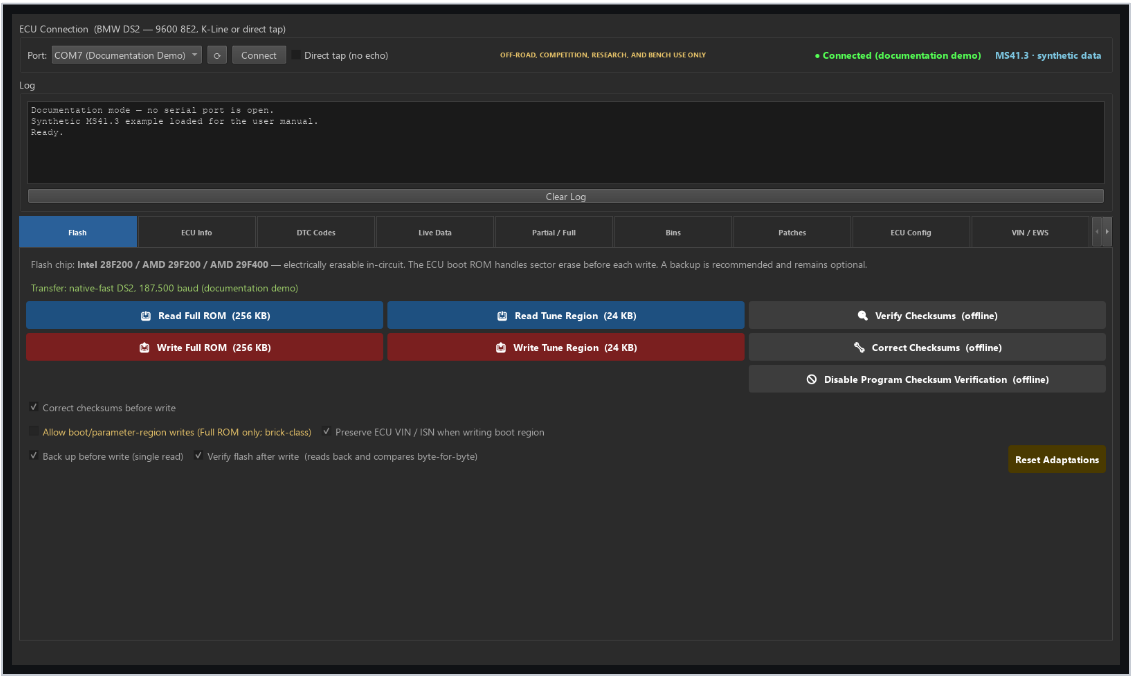
Normal K-Line versus direct tap

- Leave **Direct tap (no echo)** clear for a normal single-wire K-Line or OBD-II adapter.
- Select it only for a full-duplex ASC0 connection that does not echo transmitted bytes.
- Choose the wiring mode before connecting. It is locked while a session owns the port.
- Hardware BSL has its own COM selector because it normally uses a separate adapter and wiring.

Tab order

Tab	Primary purpose
Flash	Full and tune reads/writes through the automatically selected transfer path.
ECU Info	Live identity, firmware, calibration, VIN, chip, and loader information.
DTC Codes	Read, describe, export, and clear diagnostic trouble codes.
Live Data	Poll selected MS41 values and record CSV data.
Partial / Full	Convert between 24 KB tune and 256 KB full images.
Bins	Catalog reads, backups, generated images, and notes.
Patches	Compose, detect, migrate, or remove supported firmware patches.
ECU Config	Inspect and modify supported calibration configuration switches.
VIN / EWS	Separate VIN editing and EWS3 alignment workflows.
ROM Analyzer	Inspect a BIN offline without connecting to an ECU.
Soft-BSL	Install and operate the persistent Soft-BSL loader.
BSL-Unbricker	Hardware bootstrap reads, erase, program, and recovery.

Flash tab



Flash tab controls and transfer status

The Flash tab is the normal starting point for ECU reads and writes.

Choose the operation

- **Read Full** saves a single-pass 256 KB full ROM.
- **Read Tune** saves the 24 KB calibration region.
- **Write Full** programs the supported full-image regions while respecting boot-write controls.
- **Write Tune** programs the calibration region.

Automatic transfer selection

The application selects one of these routes:

1. **Soft-BSL** when the persistent loader is detected. It starts at the highest supported tier and retries lower Soft-BSL tiers only before erase.
2. **Native-fast DS2** on a compatible stock ECU through D2XX. A failed pre-erase high-rate check can restart the complete operation over normal DS2 only after the normal ECU state is confirmed.
3. **Normal DS2 at 9600** when neither accelerated route is available.

Fallback is allowed only before erase. Once erase may have started, the application retains the active session for recovery instead of changing transports.

Write options

- **Correct checksums before write** is enabled by default. MS41.3 boot and calibration checksums are corrected; its program checksum remains unchanged because stock program verification is disabled.
- **Back up before write** is optional and follows the operator's selection.
- **Verify flash after write** controls host-side byte-for-byte read-back verification.
- ECU-side flash finalization is independent of the optional host Verify checkbox.
- Boot/parameter writes remain separately armed because they carry a larger recovery risk.

Failure and retained-session recovery

Failure before erase

Before erase, Soft-BSL can retry the operation at a lower Soft-BSL baud tier. Native-fast DS2 can restart through normal DS2 at 9600 only after the normal low-rate ECU state has been confirmed.

Failure after erase

After erase begins, changing baud rate, reopening the port, or cycling ignition can discard the only live recovery path. The application therefore retains:

- The D2XX native-fast session for a native DS2 write failure.
- The loaded RAM agent and port ownership for a Soft-BSL write failure.
- The exact prepared target bytes required for a same-session retry.

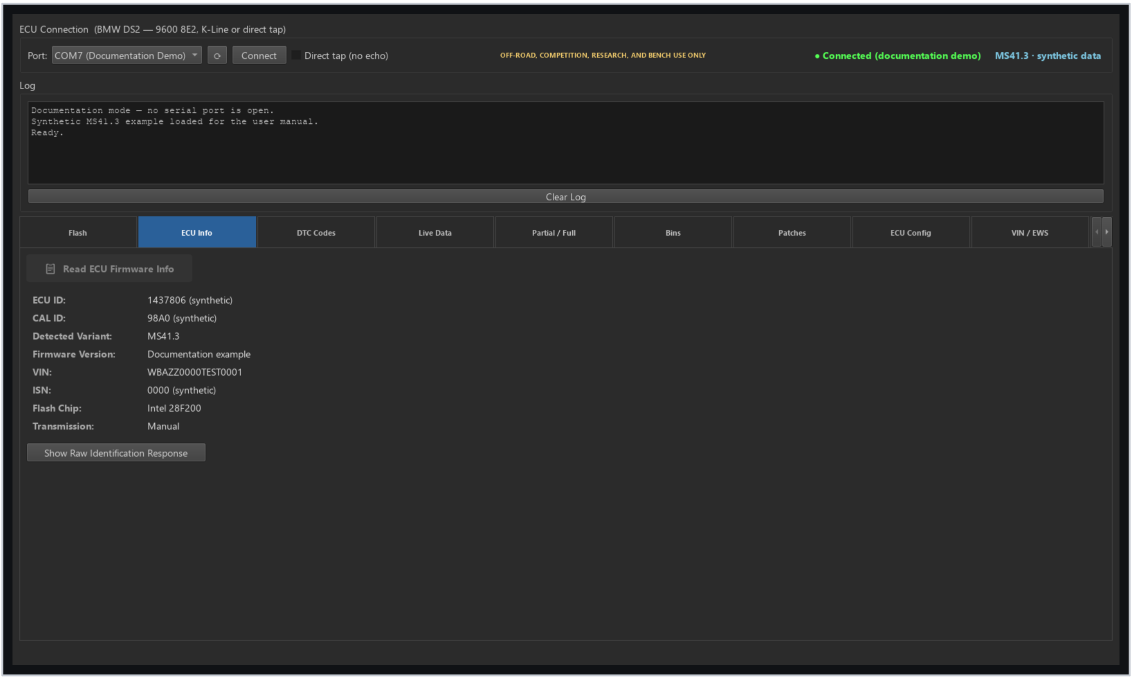
Follow the red recovery message exactly. Keep ignition ON and select **Retry Flash Recovery**.

When hardware BSL is required

Use hardware BSL when normal DS2 no longer starts and no retained Soft-BSL/native session is available. Hardware BSL runs from CPU bootstrap ROM and does not depend on valid flash contents.

> **DANGER:** Do not experiment with reset lines, ignition cycles, VPP, or chip-family selections during a retained recovery session. Preserve the session first; use hardware BSL only when the retained path is no longer available or cannot recover the target.

ECU Info, DTC Codes, and Live Data



Synthetic ECU information and diagnostics workspace

ECU Info

Use **Read ECU Information** after connecting. The tab can report the detected ECU variant, firmware/ECU ID, calibration ID, VIN, identity strings, flash-driver signature, flash-chip family, and Soft-BSL marker when available.

If MS41.2 and MS41.3 identification is ambiguous, use a full image or the stronger program and calibration markers rather than relying only on the shared ECU ID.

DTC Codes

1. Connect through normal DS2.
2. Select **Read DTCs**.
3. Review code, occurrence/state information, and the MS41 description.
4. Export the report if it is needed for service records.
5. Clear codes only after recording them and correcting the cause.

Live Data

Review the fixed parameter rows for plausible values and units. Fast telegram mode requests the mapped addresses in one batch; compatible mode reads the same mapped values in smaller blocks. CSV logging is written automatically under the application `logs/` directory while logging is enabled.

Fast telegram mode uses all 24 ECU slots. In addition to the core engine values, it reports EVAP purge duty, front-O2 and MAF input voltages, and operating states such as closed throttle, part/full load, deceleration fuel cut, and engine start. On MS41.3, the tool checks the runtime wideband flag once when polling starts. If enabled, the profile automatically reports actual AFR, target AFR, and the configured wideband input voltage; the table also identifies the selected input and whether narrowband emulation is active.

Live Data is read-only with respect to flash memory.

Offline files, conversion, and Bins

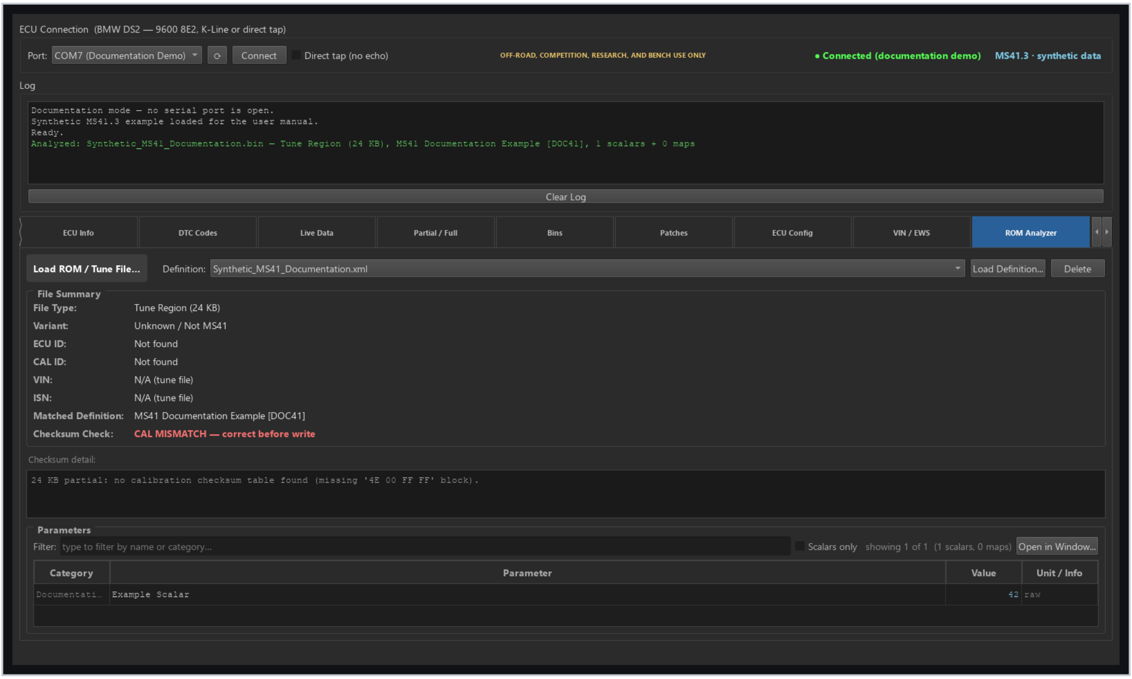
ROM Analyzer

Load a 24 KB or 256 KB BIN to inspect size, version evidence, ECU/CAL identity, VIN, checksum state, and matching definition information. No ECU connection is required.

To enable parameter matching:

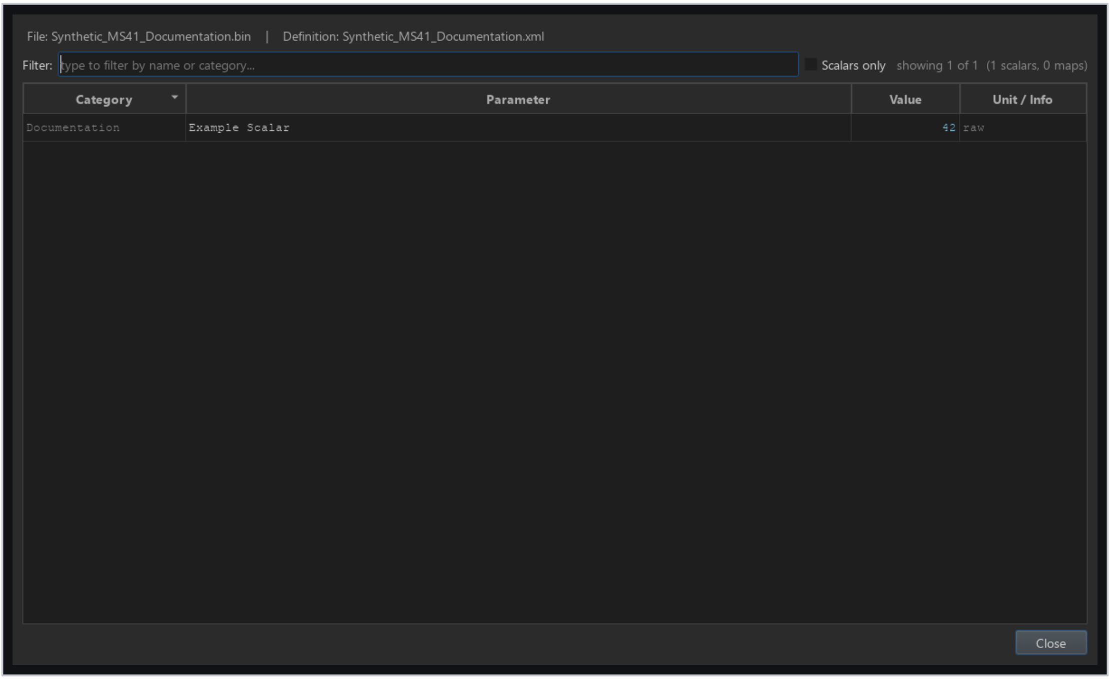
1. Select **Load Definition...** and choose a RomRaider-format MS41 XML definition.
2. The tool validates the XML and copies it into the per-user definition registry.
3. Use the **Definition** list to switch between registered definitions. The selection persists across application restarts.
4. Select **Delete** to remove only the registered copy. The original XML is never modified.

If an imported filename already exists with identical content, the existing copy is selected. If the content differs, the tool asks before replacing the registered copy. Flashing and dumping do not depend on ROM Analyzer definitions.



ROM Analyzer with a registered synthetic definition

Select **Open in Window...** in the Parameters section for a larger, resizable table. The detached window remains modeless, has its own filter and scalar-only control, supports column sorting, and updates automatically when the loaded BIN or selected definition changes.



Detached ROM Analyzer parameters window

Stop if the analyzer reports a hybrid program/calibration combination. A hybrid image can contain code and calibration from different MS41 variants and is blocked from normal flashing.

Partial / Full

- **Full to Partial** extracts the standard 24 KB tune from a 256 KB image.
- **Partial into Full** replaces the calibration in a selected full base.
- Variant conversions preserve or replace identity only according to the explicit workflow.

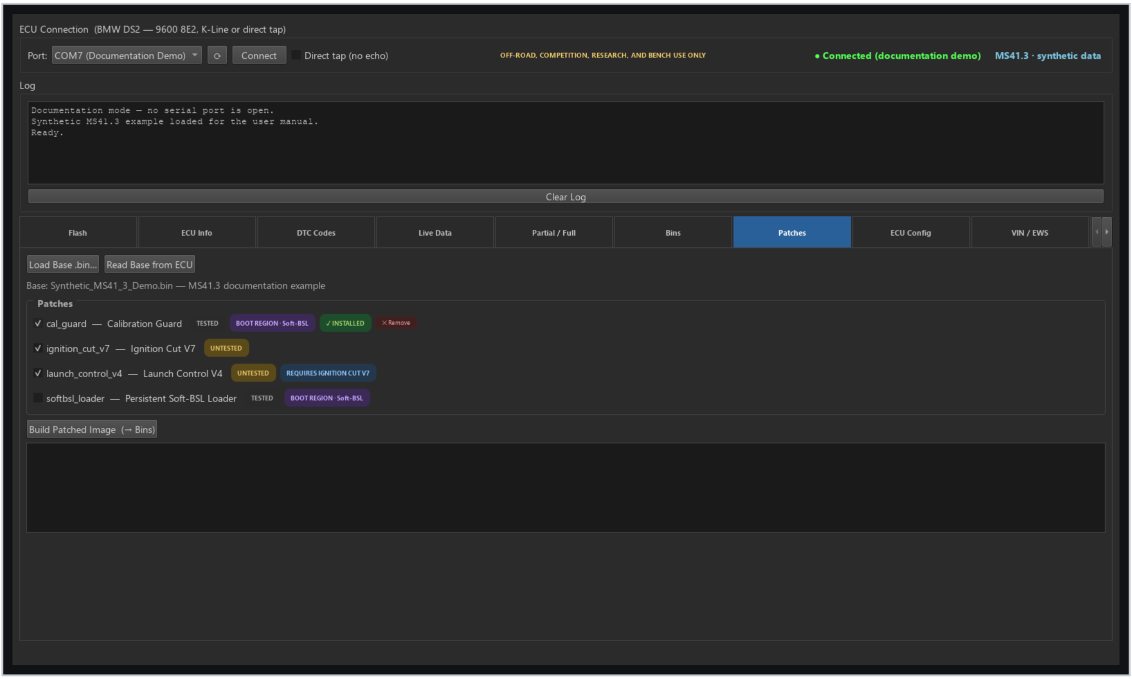
MS41.0 conversions and boot-region replacement remain guarded where hardware validation is incomplete. Read every warning before continuing.

Bins

Bins catalogs files created by reads, backups, and patch composition. Entries include available variant, type, VIN/CAL metadata, notes, and source. Use descriptive notes and preserve a known-good original separately from edited or patched images.

Select a 24 KB tune or 256 KB full-ROM entry and choose **Open in BSL-Unbricker** to load it as the hardware-BSL reference image and open that tab. This prepares the recovery controls only; it does not connect to hardware, review or approve a flash plan, or flash the ECU. A tune selects the **tune** region when available. A full ROM leaves the chip, physical half, and region unchanged for the operator to select explicitly.

Firmware patches



Patch selection with untested status

The Patches tab loads a compatible base image, checks which patches apply, detects already-installed and deprecated revisions, validates dependencies and byte collisions, recomputes required checksums, and archives the composed image into Bins.

Status and migration

- **Installed** means the patch signature is present in the loaded image.
- **Deprecated - remove only** identifies a historical revision retained for safe detection and removal. Deprecated descriptors are not offered for a new installation.
- **Untested** means emulator validation exists but physical vehicle testing has not been completed.
- Boot-region patches require a transfer path that can actually deliver their bytes.

Ignition Cut V7 and Launch Control V4 ignition mode intentionally remain marked **Untested**. Launch Control V4 fuel mode has held its configured 4000 RPM setpoint during vehicle testing. Field-failed Ignition Cut V6 remains visible only when installed so it can be removed before V7 is applied. Applying one requires an explicit confirmation. Do not treat emulator verification as proof of safe behavior on an engine.

Required external definitions

Installing or removing these firmware patches does not provide the corresponding tuning definitions. To configure Ignition Cut or Launch Control parameters in RomRaider, source a compatible definition separately and verify that it matches the ECU variant, calibration ID, and installed patch revision. Corresponding ready-to-use Ignition Cut and Launch Control definitions are not bundled with the Windows release. A mismatched definition can expose incorrect tables or write to the wrong calibration addresses.

Safe composition

1. Load or read a compatible base image.
2. Remove any detected predecessor when instructed.
3. Select the required current patches and dependencies.
4. Review boot-region and Untested badges.
5. Build the image and inspect the build log.
6. Flash the archived result through the normal Flash workflow only after reviewing it.

ECU Config and VIN / EWS

ECU Config

ECU Config exposes supported calibration configuration bits in either file mode or connected-ECU mode. Read the current values first, change only understood options, and write through the normal guarded path. Some options affect checksum verification or core engine features; their warnings are not interchangeable with the optional host read-back Verify checkbox.

Oxygen-feedback choices are resolved from the exact CAL ID because ID41, ID42, ID59, and ID85 do not share one Byte 6 encoding. The ID12/ID60 O2 disable remains experimental and is available only for a full-ROM target. Select **O2 Feedback Program Gate = Feedback Disabled** first; the calibration section will then expose **Oxygen Sensors = Disabled (Experimental)**. Switching the program gate back to enabled removes that calibration choice.

In connected-ECU mode, **Read from ECU** enables the supported program controls. If the current connection already has an archived full read, the application reuses it. Otherwise, a program change makes **Write to ECU** read and archive the ECU's unmodified 256 KB ROM before applying the selected configuration. Calibration-only changes retain the faster 24 KB partial-write route. Program changes are passed to the same guarded full-ROM writer used by the Flash tab, including automatic Soft-BSL/native-fast/standard-DS2 selection and fallback, checksum and flash-family checks, boot-region policy, confirmation, and optional read-back verification. The configuration workflow never uses a different-variant base image as a firmware conversion.

VIN editing

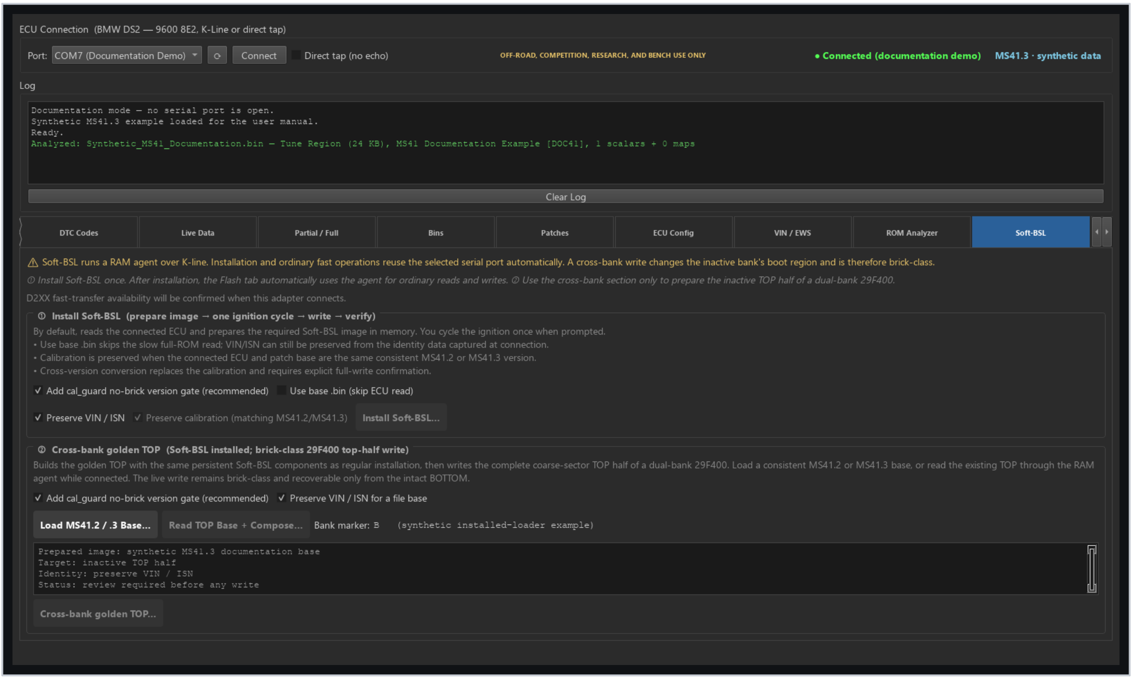
VIN editing is a Soft-BSL sector read-modify-write workflow. It requires a compatible installed loader, detected flash family, D2XX connection, and a live identity-sector cache tied to the current connection. The application changes only the packed VIN field and preserves the rest of the owning erase sector.

EWS3 alignment

EWS alignment is separate from VIN editing. It reads a fresh live DME ISN, applies the validated EWS3 encoding, rechecks the value immediately before transmission, and requires the expected EWS acknowledgement.

> **WARNING:** Do not assume that a cached VIN/identity read proves the current EWS state. VIN and EWS use separate live workflows and separate ownership checks.

Soft-BSL



Soft-BSL installation and recovery controls

Soft-BSL installs a small persistent loader in ECU flash. The loader accepts an agent over DS2, loads it into RAM, and transfers control to that agent for higher-speed reads and writes.

Installation

Use the guided installer. It identifies the target, prepares the temporary entry path, validates the current image and flash family, installs the persistent loader, and confirms normal DS2 communication after the required ignition sequence.

If the temporary Phase 1 write cannot observe the ECU-owned `E659=0xCC` readiness marker, the application closes and releases the adapter before asking for an ignition cycle and an explicit **Retry** or **Cancel**. Cancellation at this prompt is pre-erase: no challenge, selector, erase, or flash command was sent. Each repeated marker timeout requires a new decision; the retry reopens a fresh native-fast transport and never silently changes to the legacy writer.

The temporary installation entry is removed after the persistent loader is written. Subsequent daily operations use the persistent loader and current RAM agents.

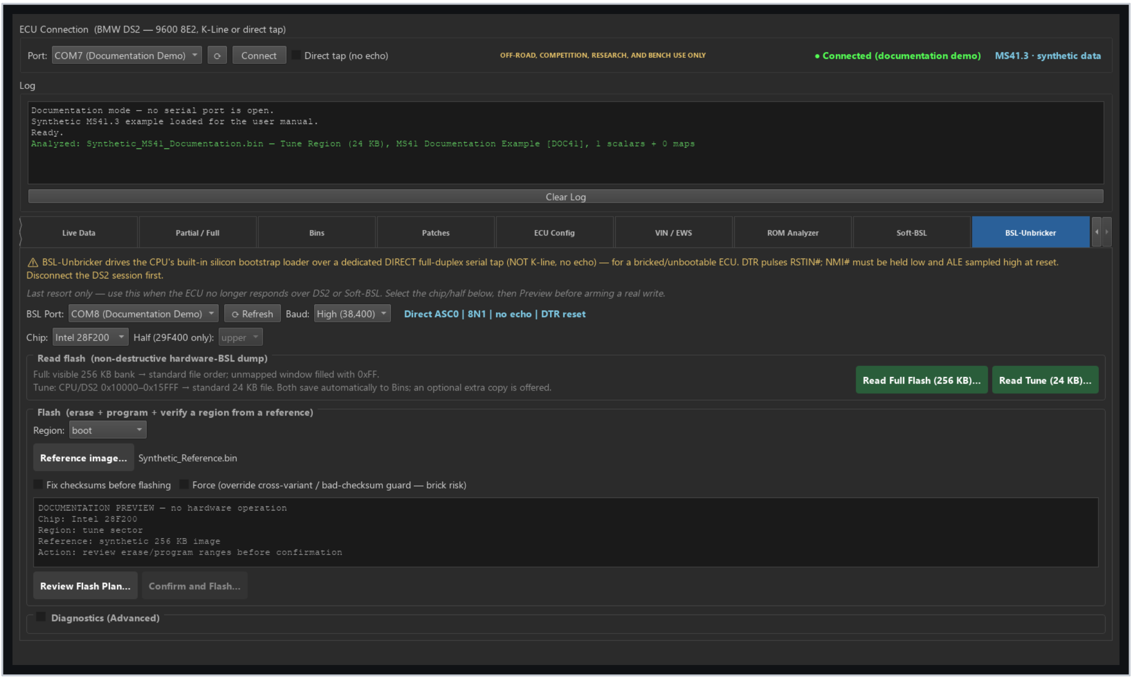
Normal operation

- Full and tune reads use the same optimized RAM-agent path.
- Intel and AMD/JEDEC devices use their matching agent and command set.
- Program writes on MS41.2 include the checksum storage required by its enabled program CRC.
- Successful writes finalize the marker independently of optional host read-back verification.

Cross-bank and boot writes

Golden-bank and boot-region workflows are advanced, recovery-sensitive operations. Follow the displayed A17/bank instructions and verify the selected physical half. Do not use them as a routine replacement for normal tune or program writes.

BSL-Unbricker



Hardware BSL review and programming controls

Hardware BSL communicates through the 80C166 bootstrap loader even when flash contents are blank or corrupt.

Required connection

- Separate full-duplex ASC0 Tx/Rx direct tap.
- 8N1 serial framing with no K-Line echo removal.
- DTR-controlled reset pulse when wired to the supported reset circuit.
- Correct ALE/NMI bootstrap-entry straps.
- Correct chip selection and 29F400 half.

Selectable rates are 9,600, 19,200, and 38,400 baud. Use 38,400 as the preferred D2XX rate and select 19,200 or 9,600 when adapter, wiring, or signal stability requires a fallback.

Intel VPP

Intel 28F200 erase/program requires the correct 12 V VPP/RP# supply. AMD/JEDEC parts do not. The VPP control remains disabled until the Intel chip family is selected.

Read and write workflow

1. Select the dedicated BSL COM port, chip, physical half, region, and rate. The application uses the DTR reset line for the supported entry circuit.
2. For a read, choose full 256 KB file order or the standard 24 KB tune.
3. For a write, select the reference image directly or prepare one from **Bins**, then choose **Review Flash Plan**.
4. Verify every physical address range and prerequisite in the preview.
5. Choose **Confirm and Flash** only when the plan is correct.
6. Monitor erase, programming, and complete physical-region read-back.
7. Remove VPP and bootstrap straps before returning to normal K-Line operation.

Troubleshooting

The COM port is unavailable

- Close other software using the adapter.
- Refresh the port list.
- Confirm whether the normal K-Line adapter or the separate BSL adapter is required.
- Reconnect the FTDI device and verify its Windows driver.

D2XX is unavailable

- Confirm the selected COM port belongs to the intended FTDI device.
- Verify the FTDI D2XX driver is installed.
- Normal DS2 can use the pyserial fallback at 9600.
- Hardware BSL can use its compatible serial fallback. Native-fast DS2 requires D2XX; Soft-BSL can fall back to its supported low-rate serial tier when D2XX is unavailable.

The fast-path stability check fails

If a Soft-BSL tier fails before erase, the application can retry a lower Soft-BSL tier. If a native-fast DS2 check fails before erase and normal ECU state is confirmed, it can restart the complete operation over normal DS2. Check adapter latency, wiring, ground, ECU voltage, and signal integrity before retrying.

A write failed

- If the UI says erase did not start, correct the reported connection or file problem and retry.
- If the UI says recovery is active, keep ignition ON and use **Retry Flash Recovery**.
- If the ECU no longer boots and no retained session exists, prepare the hardware-BSL connection.

The ROM Analyzer cannot load definitions

Use **Load Definition...** to select a valid RomRaider-format MS41 XML file. Do not copy XML files into `_internal` or any other packaged runtime directory. If a registered definition was changed or damaged outside the application, delete it and import a known-good copy again. Confirm the BIN size and exact ECU software identity before relying on matched values.

Values or identity look wrong

Stop the operation. Confirm the connected ECU, file, variant, byte order, and definition. Re-read the value through an independent path before writing anything.

Final checklists

Before a normal read

[ ] ECU voltage is stable.

[ ] The normal K-Line/direct-tap selection matches the wiring.

[ ] The correct COM port is selected.

[ ] Full 256 KB or tune 24 KB is intentionally selected.

[ ] The output location does not overwrite an irreplaceable file.

Before a normal write

[ ] The target file size and ECU variant match the intended job.

[ ] The detected flash-chip family is compatible with the image and path.

[ ] Checksum correction is enabled unless there is a documented reason otherwise.

[ ] Backup and Verify choices reflect the operator's decision.

[ ] Stable power and hardware recovery are available.

[ ] No other software owns the COM port.

Before hardware BSL programming

[ ] Direct ASC0 wiring and bootstrap straps are correct.

[ ] Chip family and 29F400 half are correct.

[ ] Intel VPP is active only for the Intel write phase.

[ ] The reviewed physical address plan matches the intended region.

[ ] The reference image is the correct size and byte order.

[ ] A recovery plan exists if programming is interrupted.

Disclaimer

BimmerStein ECU Tool is experimental software intended solely for off-road, competition, research, and bench use. It is provided "as is," without warranty of any kind, to the maximum extent permitted by applicable law.

ECU programming, calibration changes, firmware patches, and recovery operations can cause data loss, an unbootable ECU, engine or vehicle damage, unexpected engine behavior, or unsafe operating conditions. The user assumes all risks associated with connecting, configuring, modifying, or flashing an ECU and is responsible for maintaining suitable backups, stable power, and an appropriate recovery method.

The user is solely responsible for determining whether any operation or modification is legal and compliant with applicable emissions, safety, registration, competition, and other regulations. An off-road designation does not establish that a particular modification is lawful.

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